No Genuine Copula in Japanese: the pseudo-copula =*da* as an existential, conjunctive and locative complex[1]

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Abstract:

This paper proposes an alternative analysis of the Japanese so-called copula =*da* and its non-contracted form =*de aru*. While =*da* has traditionally been treated as a copula equivalent to English *be*, we argue instead that the nature of this pseudo-copula can be best understood in terms of **how Japanese, which lacks the primitive *be*, employs its most basic linguistic resources — namely, the locative postposition =*ni*, the conjunctive suffix -*te* and the existential verb *ar*-, which together still autonomously underlie =*da* — in denoting relations between two entities**.

Under this conception, the sentence *A*=*wa B*=*de aru* 'A is B' is semantically analysed as both existence-introducing and existence-domain-specifying. Syntactically, it consists of the one-argument obligatory construction [*A*=*wa aru*] 'A exists' and the adjunct [*B*=*de*] 'with (A's existence) located in the conceptual domain of B' or, more practically, 'and it is as B'. Schematically, this construction can be represented as the conjunctive proposition EXIST(A) ∧ IN(rEXIST(A), B), distinct from the atomic proposition BE(A, B). Although *B*=*de* is syntactically adjunctive, it cannot be omitted because it is information-structurally indispensable: it marks the informational focus, whereas the syntactically obligatory *A*=*wa aru* provides the presuppositional background.

This reanalysis makes possible a unified account of the structure, semantics and diachronic development of Japanese so-called copular constructions.

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1. Introduction — morphosyntactic and prosodic separability of *de* and *aru* in *dearu* and syntactic and prosodic constituency of *de* and the preceding NPs

In Japanese linguistics, *da/dearu* has traditionally been treated as a copula equivalent to *to be* in English, as in (1), where *dearu* can be replaced by *=da*, '=' indicating enclisis:

(1)	kare=wa	gakusei	dearu
	he(SUBJ)=TOP	student(PRED)	COP
	'He is a student.'		

However, this analysis faces a major challenge: it cannot account for the **high degree of independence between *de* and *aru* in *dearu***. For instance, when the noun phrase (NP) *gakusei* 'student' is marked by the focus/additive enclitic particle *=mo* 'also' — one of the so-called *toritatesi* in Japanese linguistics — the sentence cannot take the expected form shown in (2):

(2)	*kare=wa	gakusei	=mo	**dearu**
	he(SUBJ)=TOP	student(PRED)	also(ADD)	COP
	'He is also a student.'			

but must be as in (3):

(3)	kare=wa	gakusei	**de**	=mo	**aru**
	he(SUBJ)=TOP	student(PRED)	COP?	also(ADD)	COP?
	'He is also a student.'				

The particle *=mo* is inserted between *de* and *aru*, as if behaving as an infix from the perspective that treats *dearu* as a copula — a situation comparable to the adverb *also* being inserted into the copula *is* in English, as in **he i- also -s a student*. Furthermore, in (4):

(4)	kare=wa	gakusei	**deari**	kanozyo=wa	syakaizin	**=da**
	he(SUBJ)=TOP	student(PRED)	COP	she(SUBJ)=TOP	working adult(PRED)	COP
	'He is a student, she is a working adult.'					

the first *deari* may be reduced to *de* alone, without *ari* — formally, the conjunctive form, or *ren'yōkei*, of the verb *aru* — as in (5):

(5)	kare=wa	gakusei	**de**	kanozyo=wa	syakaizin	**=da**
	he(SUBJ)=TOP	student(PRED)	COP?	she(SUBJ)=TOP	working adult(PRED)	COP
	'He is a student, she is a working adult.'					

Prosodically, too, the so-called copula *dearu* exhibits unexpected behaviour. If *dearu* were a single word, it would be expected to constitute either an independent prosodic word (ω) or an enclitic, so the utterance *kare=wa gakusei dearu* would be expected to be prosodised as either (6) or (7), where " | " marks a prosodic-word boundary, and prosodic words in Japanese generally correspond to traditional *bunsetu*:

(6) | *kare=wa* | *gakusei***=*dearu*** | — *dearu* as an enclitic (i.e. encliticised to *gakusei*)

(7) | *kare=wa* | *gakusei* | ***dearu*** | — *dearu* as an independent ω

In actual speech, nevertheless, the prosodisation is implemented as in (8):

(8) | *kare=wa* | *gakusei=****de* | *aru*** | — *de* as an enclitic and *aru* as an independent ω

Since prosodic-word boundaries are determined with reference to morphosyntactic information (Nespor & Vogel 2007), the fact that the utterance is prosodised as | *kare=wa* | *gakusei=de* | *aru* | strongly suggests the presence of a morphosyntactic boundary between *de* and *aru* — namely, that *de* and *aru* constitute distinct morphosyntactic elements. This prosodic separation is further supported by the fact that the discourse enclitic particle *=ne* — one of the so-called *syūzyosi* in Japanese linguistics — can intervene between *de* and *aru*, as in (9), which would be impossible if *dearu* constituted a single prosodic word:

(9) | *kare=wa=ne* | *gakusei=****de****=ne* | ***aru****=ne* |

All of these facts demonstrate that ***de* and *aru* are morphosyntactically and prosodically separable**, whereas **the NP *gakusei* and *de* are always syntactically and prosodically inseparable**, as illustrated in (10):

(10) *Kare=wa* [*gakusei de*] [*aru*].

Kare=wa [*gakusei de*] *mo* [*aru*].

Kare=wa [*gakusei de*] [*ari*], *kanozyo=wa syakaizin=da.*

Kare=wa [*gakusei de*], *kanozyo=wa syakaizin=da.*

Kare=wa=ne [*gakusei de*]*=ne* [*aru*]*=ne.*

Hence, in *kare=wa gakusei dearu*, the sequence *gakusei=de* is assumed to constitute a syntactically and prosodically defined constituent, as Dalla Chiesa (2015) notes: "the NP-*de* element must always be the phrase (in the closest proximity to the verb)"[2], where the parenthetical remark is by the present author. This *gakusei de* is formally identical to a **postpositional phrase (PP) consisting of a NP combined with the so-called postposition =*de***. In what follows, hence, *gakusei de* will be treated provisionally as a PP, that is, the PP *gakusei=de*. This separation between *=de* and *aru* is compatible with Matsuoka's observation that "the copula *da* consists of the postposition *de* and the verb *aru*" (2019: 142). The fact that *gakusei=de* and *aru* each form separate prosodic words can thus be naturally accounted for.

2. NP + *=de* as a conceptual-locative adjunct ConjP and *aru* as a one-argument existential verb

In Japanese, **PPs of the form [NP + =*de*] generally function as adjuncts**, as Hasegawa (1999: 26, translated from Japanese) states: "The case particle *=de* occurs not only for 'location of an action' but also for a variety of other elements such as 'cause' (*kaze=de yasumu* 'to be absent because of a cold'), 'means' (*genkin=de kau* 'to buy with cash'), 'time' (*mikka=de tukuru* 'to make in three days') and 'state' (*hadaka=de odoru* 'to dance naked'), and all of these fundamentally function only as adjuncts. There are no predicates that require these as arguments indispensable for their informational completeness. From this, one can generalise that *=de* marks adjuncts." Based on this generalisation, *kare=wa gakusei=de aru* can be syntactically analysed as consisting of the **obligatory structure [*kare=wa aru*]** and the **adjunctive structure [*gakusei=de*]**. Further justification for treating *gakusei=de* not as an argument but rather as an adjunct will be provided below in this section and in Section 8.

In *kare=wa aru*, the verb *aru* — or, more precisely, its stem *ar-* — can thus be analysed as a **ONE-ARGUMENT predicate**, taking only the NP *kare* as its argument. When the verb ***aru* functions as a one-argument predicate**, it means '**to exist**', as in *kanōsei=wa* ***aru*** 'the

[2] She argues that the ungrammaticality of **neko=**_**de**_ *wagahai=ga* ***aru*** 'I am a cat' supports her claim that "the NP-*de* element must always be the phrase in the closest proximity to the verb". However, the ungrammaticality of this sentence does not arise from word order, but rather from the fact that *wagahai*, while functioning as a topic, is not marked by the particle *=wa*. Indeed, although fairly awkward, *neko=****de*** *wagahai=****wa*** ***aru*** is not completely ungrammatical, and expressions such as NP=***de*** *watasi=****wa*** ***ari****-tai* 'as the referent denoted by the NP I want to exist' — e.g. *ryōsikitekina syakaizin=****de*** *watasi=****wa*** ***ari****-tai* 'as a rational member of society I want to exist' — sound much more natural.

possibility exists'. The sentence *kare=wa aru* can thus be interpreted as 'he exists', **introducing the existence** of *kare*. (In Contemporary Japanese, however, this use of *aru* as a one-argument existential verb often has a somewhat literary or philosophical flavour, as in *ware omou, yueni ware* ***ari*** 'I think, therefore I am' or "***aru***"*=to iu koto=wa dō iu koto=ka=to iu toi* 'the problem of what it means "to exist"'. The verb *aru* is more commonly used as a two-argument existential or locative verb as in *tukue=no ue=ni hon=ga aru* 'there is a book on the table' or *hon=wa tukue=no ue=ni aru* 'the book is on the table'. In everyday registers, the verb *sonzai-suru* is often preferred for expressing one-argument existential predication.)

This meaning is reflected in the commonly used compound nouns ***ari****kata* and ***ari****yō*, which signify 'mode/manner of existing, ideal way to exist' and 'state/appearance of existing', respectively. Both are derived from the stem *ari-* of the verb *aru* in this sense — namely, 'to exist' — combined with the nominal formatives *-kata* 'mode, manner, way' and *-yō* 'state, appearance', respectively. Further justification for analysing *aru* in this construction as 'to exist' — a meaning that, for pragmatic and informational reasons, is rarely brought to the speaker's or hearer's awareness in ordinary categorical or identificational contexts of the type 'A = B' or, more precisely, 'A ∈ B' — will be presented in Section 6.

As for the PP *gakusei=de*, on the other hand, although **PPs headed by** ***=de*** can denote, as mentioned above with reference to Hasegawa (1999: 26), a range of semantic roles such as location, cause, means, time or state, the most plausible interpretation in the present construction is **locative**, namely the primitive sense of *=de*. Crucially, however, this use does not denote a physical location but a conceptual location. Here, conceptual location refers to the localisation or anchoring of an entity within a conceptual space structured by domains mutually defined through semiological opposition and founded on an analogy with physical space. Consequently, *gakusei=de* means 'located in the **conceptual domain** of "student"' or, more practically, 'as a student'.

A conceptual domain is a **category construed as a locus for locating an existence**. A category here is a **conceptual grouping of entities sharing common features as constructed by human cognitive processes**. Put simply, a category exists only as a construct of the human mind. Whether a category contains a single member or multiple members is irrelevant. An entity here is understood as a constituent of all kinds of worlds, actual or not, rather than as a spatiotemporally instantiated object. Even when *B* is a proper noun in the construction *A=wa B=de aru*, *B* denotes a conceptual domain mapped from a category defined by the feature of being referable by that proper name. Likewise, *kimi* 'you' in *watasi=ni=totte itiban taisetuna=no=wa kimi=da* 'the most important thing for me is you' is understood as a conceptual domain defined by the category 'the unique addressee in the context of utterance or the speech situation'. For instance, the construction in (11):

(11) *Kare=wa syakaizin**=de** aru: sigatu=ni gakusei**=kara** syakaizin**=ni** natta.*
'He is a working adult: in April, he moved from being a student to being a working adult.'
(lit.) He exists **in** the conceptual domain of "working adult": in April, he moved **from** the conceptual domain of "student" **to** that of "working adult".

indicates that the same entity denoted by *kare* has moved within the conceptual space **from** the domain *gakusei* 'student' **to** the distinct domain *syakaizin* 'working adult', thereby relocating its existence **in** the latter domain. The postpositions *=de* 'in' in *gakusei=de*, *=kara* 'from' in *gakusei=kara* and *=ni* 'to' in *syakaizin=ni* are plausibly understood as **metaphorical extensions of the physical location, source and goal markers**, as in the construction in (12):

(12) *Kare=wa Tōkyō**=de** hataraite=iru: sigatu=ni Kyōto**=kara** Tōkyō**=ni** yatte=kita.*
'He is working in Tokyo: in April, he came to Tokyo from Kyoto.'
(lit.) He is working **in** the physical domain of Tokyo: in April, he came **from** the physical domain of Kyoto to that of Tokyo .

Given that constructions encoding conceptual space systematically copy the constructions encoding physical space — that is, the linguistic isomorphism between physical space and conceptual space — it can be argued that it is not that speakers intentionally choose a spatial metaphor, but rather that the grammar provides no alternative formal resources, and consequently that Japanese grammar appears to formally force conceptual category domains to be treated in spatial terms.

Syntactically, the core phrasal structure found in *kare=wa gakusei=de aru* can be analysed provisionally as [$_{VP_1}$ [$_{PP}$ *gakusei=de*] [$_{VP_0}$ *kare ar-*]]. In the lower VP [$_{VP_0}$ *kare ar-*] 'he exists', the stem *ar-* of the existential verb *aru* and the NP *kare* constitute a one-argument predicate and its theme argument, respectively. In the higher VP [$_{VP_1}$ [$_{PP}$ *gakusei=de*] [$_{VP_0}$ *kare ar-*]] 'he exists, with his existence located in the conceptual domain of "student"', on the other hand, the postposition *=de* — or, more precisely, **the underlying postposition =*ni* contained within the surface postposition–conjunction complex =*de***, as will be discussed below — **behaves as if it were a two-argument predicate, and locates the lower VP [$_{VP_0}$ *kare ar-*] 'he exists',**

construed as a reified eventuality 'his existence', within the conceptual domain denoted by the adjunctive NP [*gakusei*] 'student'. Semantically, therefore, the construction expresses 'He exists, with his existence located in the conceptual domain of "student"' or, more practically, 'He exists, and it is as a student'. Since the adjunct *gakusei=de* — containing the predicate-like *=ni* within *=de* — is introduced precisely for such a predication, **the most crucial logical** — though not syntactic — **predicate in this construction is** not the verbal stem *ar-* — wihch introduces *kare*'s existence in the lower VP [$_{\mathrm{VP}_0}$ *kare* ***ar-***] — but rather **the predicate-like postposition *=ni*** contained within the postposition–conjunction complex *=de* — which, in the higher VP [$_{\mathrm{VP}_1}$ [$_{\mathrm{PP}}$ *gakusei=**de***] [$_{\mathrm{VP}_0}$ *kare ar-*]], locates *kare*'s existence denoted by the lower VP [$_{\mathrm{VP}_0}$ *kare ar-*] in the conceptual domain referred to by the NP [*gakusei*].

Schematically, the more abstract construction ***A=wa B=de aru*** — in which *A=wa aru* 'A exists', even without *B=de* 'with (A's existence) located in the conceptual domain of B', constitutes an atomic proposition, since *B=de* is assumed here to be adjunctive — can be represented as **E(A) ∧ IN(rE(A), B)**. Here the first proposition E(A) 'A exists' — where E abbreviates 'exist' — denotes **the introduction of existence**, whereas the other proposition IN(rE(A), B) 'A's existence is located in the conceptual domain of B' represents **the specification of the domain of existence**. Thus, *A=wa B=de aru*, representable as E(A) ∧ IN(rE(A), B), means 'A exist, with A's existence located in the conceptual domain of B' or, paraphrased, 'A exists — that is, A is a constituent of all kinds of worlds, actual or not —and the question of where A exists conceptually — but not physically — is answered by locating A's existence in the conceptual domain B.'

The function *r* in rE(A) — which abbreviates 'reified' — maps the proposition E(A) 'A exists' to the corresponding reified eventuality rE(A), that is, an eventive entity representing 'A's existence'. This analysis requires a three-tier mapping:

(i) from the proposition E(A) 'A exists';

(ii) to the eventuality corresponding to E(A) in the relevant world (whether actual or not);

(iii) to the reified eventuality rE(A) corresponding to 'A's existence', which functions as an entity capable of serving as *x* in the relational predication IN(x, B), headed by the predicate-like *=ni* within the postposition–conjunction complex *=de*.

The structure $E(A) \wedge IN(rE(A), B)$ thus constitutes a conjunctive proposition. Note that, since $IN(rE(A), B) \Rightarrow E(A)$, it follows that $E(A) \wedge IN(rE(A), B) \equiv IN(rE(A), B)$; hence **$E(A) \wedge IN(rE(A), B)$ is logically redundant**, although it is still assumed to be **indispensable for a faithful formalisation of the syntactic–semantic structure**.

Let us return to the adjunctivity of the PP *B=de* in *A=wa B=de aru*. As will be discussed in Section 7, the contemporary postposition *=de* originated historically from the sequence *=ni-te*, where *=ni* is assumed to have functioned as a locative postposition and *-te* as a conjunctive suffix. The development was probably driven by the deletion of *-i-* and by postnasal voicing — typical of the Yamato lexical stratum — yielding the sequence *=ni-te* → *=n-te* → *=n-de* → *=de*. Even today, in PPs of the form [NP + *=de*], one can observe stylistic variation between the postposition *=de* and the postposition–suffix sequence *=ni-te*, as in *Tōkyō=**de*** / *Tōkyō=**ni-te*** 'in Tokyo, at Tokyo', though the latter is a more formal and literal form. Given this diachronic development and the synchronic variation, it is plausible to assume that the underlying form of *B=de* is still *B=ni-te*, where, strictly speaking, ***B=ni* denotes conceptual location** and **the conjunctive suffix *-te* adjoins *B=ni*, not as an argument but as an adjunct**, to *A=wa aru* — or more precisely, **to the VP node [*A ar-*] within *A=wa aru***.

An NP marked by *=ni* functioning as an argument is required by the predicate as an element semantically indispensable for completing the predicate's meaning. The link between the predicate and such an argumental *NP=ni* is determined by the argument structure inherent in the predicate, and is therefore intrinsic and necessary to it. By contrast, an underlying *NP=ni* functioning as an adjunct is optional for the predicate or for some node in the predicate phrase, and the link between the adjunct *NP=ni* and the predicate or such a node is extrinsic and contingent to the predicate. In Contemporary Japanese, this extrinsic and contingent relation needs to be made explicit in the linguistic form, and the device used for this purpose is the conjunctive suffix *-te*. It may be assumed, from a cognitive perspective, that the *conjunctivity* of *-te* has been metaphorically extended — via the schema of *junctivity* — into *adjunctivity*, as schematised in (13):

junctivity

(13) metaphor: **conjunctivity** ------> **adjunctivity**

Having thus developed into an adjunctive suffix, *-te* adjoins *NP=ni* to the predicate or to some node in the predicate phrase as an adjunct rather than as an argument. The surface realisation of this underlying *NP=**ni-te*** is *NP=**de***. Thus, the use of *=de* as an adjunct marker in Contemporary Japanese can be accounted for by the fact that its underlying form *=ni-te* contains the conjunctive suffix *-te*, which functions adjunctively in this context. Consequently, *B=ni-te*, surfacing as *B=de*, can be analysed not as a PP but as a conjunction phrase (ConjP) headed by the conjunction *-te* with the complement PP *B=ni*, as illustrated in (14):

(14) $[_{\text{ConjP}}\ [_{\text{PP}}\ [_{\text{NP}}\ B]\ [_{\text{P}}\ =ni]]\ [_{\text{Conj}}\ \text{-}te]]$

In the construction *A=wa B=**de** aru*, therefore, it seems accurate to characterise *=de* as a **complex consisting of the predicate-like conceptual-locative postposition *=ni* and the adjunctive-conjunction suffix *-te***.

The remaining problem is to determine what element in the linguistic form *A=wa B=de aru* formally corresponds to the operator ∧, which expresses logical conjunction in the proposition E(A) ∧ IN(rE(A), B). The phrasal structure corresponding to the schematic representation E(A) ∧ IN(rE(A), B) is assumed to be as in (15), in line with the analysis in (14):

(15) $[_{\text{VP}_1}\ [_{\text{ConjP}}\ B{=}ni\text{-}te]\ [_{\text{VP}_0}\ A\ ar\text{-}]]$

where the lower VP $[_{\text{VP}_0}\ A\ ar\text{-}]$ corresponds to the proposition E(A), and the higher VP $[_{\text{VP}_1}\ [_{\text{ConjP}}\ B{=}ni\text{-}te]\ [_{\text{VP}_0}\ A\ ar\text{-}]]$ — formed by adjoining the adjunct PP $[_{\text{PP}}\ B{=}ni]$ to the lower VP $[_{\text{VP}_0}\ A\ ar\text{-}]$ via the adjunctive-conjunction *-te* — corresponds to the conjunctive proposition E(A) ∧ IN(rE(A), B). The semantic composition corresponding to the structure in (15) is given in (16):

(16)

$[_{\text{NP}}\ A]$:	A
$[_{\text{V}}\ ar\text{-}]$:	E( )
$[_{\text{VP}_0}\ A\ ar\text{-}]$:	E(A)
$[_{\text{NP}}\ B]$:	B
$[_{\text{P}}\ =ni]$:	IN(r( ),)
$[_{\text{Conj}}\ \text{-}te]$:	∧
$[_{\text{PP}}\ B{=}ni]$:	IN(r( ), B)
$[_{\text{ConjP}}\ B{=}ni\text{-}te]$:	∧ IN(r( ), B)
$[_{\text{VP}_1}\ B{=}ni\text{-}te\ A\ ar\text{-}]$:	E(A) ∧ IN(rE(A), B)

As stated above, E(A) is the proposition 'A exists'; rE(A) is the actual eventuality that makes this proposition true and, when reified, yields the eventive entity 'A's existence'; IN(rE(A), B) is the other proposition 'A's existence is located in the conceptual domain of B'; and what corresponds to the logical conjunction operator ⋀, which combines these two propositions, is the adjunctive-conjunction *-te*. Accordingly, the conjunction *-te* both syntactically adjoins the PP $[_{\text{PP}}$ *B=ni*] to the VP $[_{\text{VP}_0}$ *A ar-*] as an adjunct, and, logically, functions as the conjunctive operator ⋀, introducing the predicate IN(r( ),), denoted by the predicate-like postposition *=ni*. Thus, the lower VP $[_{\text{VP}_0}$ *A ar-*] already constitutes an independent proposition E(A) and, by merging this lower VP with the adjunctive ConjP $[_{\text{ConjP}}$ *B=ni-te*], which expresses ⋀ IN(r( ), B), the higher VP $[_{\text{VP}_1}$ *B=ni-te A ar-*] is formed, representing the additional proposition ⋀IN (rE(A), B).

In opposition to this *A=wa* ***B=de*** *aru*, the semantically related and syntactically similar construction *A=wa* ***B=ni*** *aru/iru* 'A is located in B' — in which *A=wa aru/iru* 'A is located', without *B=ni* 'in B', cannot constitute a proposition, since *B=ni* is argumentative, as dicussed in Section 4 — can be represented as **IN(A, B)**, which denotes the **specification of the domain of location**. IN(A, B) thus constitutes an **atomic proposition**. The compositional difference between the conjunctive E(A) ∧ IN(rE(A), B) and the atomic IN(A, B) is reflected in their default negation patterns, such as the partial negation *A=wa* <u>*B=de=**wa***</u> ***nai*** in the former vs. the total negation *A=wa* <u>*B=ni*</u> ***nai/i-nai*** in the latter — namely, E(A) ⋀ ¬IN(rE(A), B) vs. ¬IN(A, B) — which will be discussed in Section 8.

Thus, instead of the traditional **copula-based** analysis in (17) = (1), representable as BE(kare, gakusei):

(17) kare=wa gakusei **dear**-u
he(SUBJ)=TOP student(PRED) **COP**-PRES
'He is a student.'

in this paper, a **no-copula-based** analysis in (18), representable as E(kare) ⋀ IN(rE(kare), gakusei), is adopted, where CONCEP-LOC and CONJ abbreviate 'conceptual-locative' and 'conjunctive', respectively:

(18) kare=wa gakusei=**de** **ar**-u
he(SUBJ)=TOP student=**CONCEP-LOC–CONJ (ADJUNCT)** **exist**-PRES
'He exists, with his existence located in the conceptual domain of "student".'

The three fundamental conceptual devices employed in Japanese to denote relations between two entities are:

- **Predicate-like conceptual-locative postposition =*ni* 'in'**;
- **Adjunctive-conjunction suffix -*te* 'and'**; and
- **One-argument existential verb *ar-* 'exist'**.

3. The so-called postposition =*de* as an Eventuality-Domain Specifier: ∧ (r( ), NP)

In the previous section, the constituent *gakusei=de* in the construction *kare=wa* ***gakusei=de*** *aru* — representable as EXIST(kare) ∧ IN(rEXIST(kare), gakusei) — was analysed as the underlying ConjP ***gakusei=ni-te***, which corresponds to the logical component **∧ (r( ), gakusei)**. In this construction, the underlying postposition =*ni* functions conceptual-locatively, and the NP *gakusei* denotes the conceptual domain in which *kare*'s existence is located, as shown in (19):

(19)	kare=wa	gakusei=**de** (underlyingly gakusei=**ni-te**)	ar-u
	he(SUBJ)=TOP	student=**CONCEP-LOC–CONJ (ADJUNCT)**	exist-PRES
	'He exists, with his existence located in the conceptual domain of "student".'		

However, the adjunctive ConjP [NP + =*de*] can function not only conceptual-locatively, as in (19), but also **physical-locatively**, as shown in (20):

(20)	kare=wa	Tōkyō=**de** (underlyingly Tōkyō=**ni-te**)	hataraite=i-ru
	he(SUBJ)=TOP	Tokyo=**PHYSIC-LOC–CONJ (ADJUNCT)**	working=be-PRES
	'He is working, with his working located in the physical domain of "Tokyo".'		

Thus, the construction *kare=wa* ***Tōkyō=de*** *hataraite=iru* can be schematically represented as WORK(kare) ∧ **IN(r**WORK(kare), **Tōkyō)**, in parallel to EXIST(kare) ∧ **IN(r**EXIST(kare), **gakusei)** from *kare=wa* ***gakusei=de*** *aru*.

In both sentences, the postposition–conjunction complex =*de*, representable as **∧ IN(r( ), NP)**, ultimately maps the propositions WORK(kare) and EXIST(kare) — expressed by the obligatory constructions *kare=wa hataraite=iru* and *kare=wa aru*, respectively — to the corresponding reified eventualities rWORK(kare) and rEXIST(kare). These reified eventualities are then located by the underlying postposition =*ni* — amalgamated with -*te* to surface as =*de* — within the domains denoted by the NPs *Tōkyō* and *gakusei*, yielding additional propositions

IN(rWORK(kare), Tōkyō) and IN(rEXIST(kare), gakusei). The underlying conjunction *-te* adjoins these eventuality-domain-specifying propositions to the base propositions WORK(kare) and EXIST(kare), yielding the conjunctive propositions WORK(kare) ∧ IN(rWORK(kare), Tōkyō) and EXIST(kare) ∧ IN(rEXIST(kare), gakusei). Accordingly, it can be assumed that **the Japanese postposition–conjunction complex =*de*** does not function as a simple locative postposition but rather as what can be called an **eventuality-domain specifier: ∧ (r(   ), NP)**.

4. The differences between *sore wa gakkō=DE ARu* and *sore wa gakkō=NI ARu*

In contrast to the construction *sore=wa gakkō=**de** **aru*** in (21), hypothesised as **existence-introducing and existence-domain-specifying** in Section 2:

(21) sore=wa gakkō=**de** **ar**-u
it(SUBJ)=TOP school=**CONCEP-LOC–CONJ(ADJUNCT)** **exist**-PRES
'It exists, with its existence located in the conceptual domain of "school".'

in genuine locative — namely, **location-specifying-only** — constructions as in (22) and (23), where PHYSIC-LOC abbreviates 'physical-locative':

(22) sore=wa gakkō=**ni** **ar**-u
it(SUBJ)=TOP school=**PHYSIC-LOC(ARGUMENT)** **be located**-PRES
'It is located in the physical domain of "school".'

(23) kare=wa gakkō=**ni** **i**-ru
he(SUBJ)=TOP school= **PHYSIC-LOC(ARGUMENT)** **be located**-PRES
'He is located in the physical domain of "school".'

the **physical-locative argument is obligatory**, as mentioned in Section 2, **and always marked by the postposition =*ni***, rather than =*de*. Thus, the verb *aru* — more precisely, its stem *ar-* — when used as a **LOCATIVE verb**, is a **TWO-ARGUMENT predicate with the argument structure [physical domain, theme]**, the location being denoted by the postpositional phrase [NP + =*ni*], as in *sore=wa gakkō=**ni aru***. When the theme is an animate entity, the verb *iru* with the stem *i-* is used instead as in (23), with the **argument PP [NP + =*ni*] unchanged**, as in *kare=wa gakkō=**ni iru*** 'he is in the school'. By contrast, the sequence of words in (24), in which the argumentative PP [NP + =*ni*] in (23) has been replaced by the ConjP [NP + =*de*]:

(24) kare=wa gakkō=**de** **i**-ru
he(SUBJ)=TOP school=LOC–CONJ(**ADJUNCT**) ?-PRES
' ? '

is semantically uninterpretable — even for native speakers — and ungrammatical, indicating that the ConjP [NP + *=de*] is adjunctive in this construction, which lacks an indispensable argument headed by *=ni*.

Given this general adjunctivity of the ConjP [NP + *=de*], *kare=wa gakusei=de aru* may reasonably be regarded SYNTACTICALLY — though NOT PRAGMATICALLY OR INFORMATIONALLY, as will be discussed in Section 5 — as a **structurally self-contained, existence-introducing construction** *kare=wa aru* 'he exists', **accompanied by the conceptual-locative adjunct** *gakusei=de* 'with (his existence) located in the conceptual domain of "student"', **which conjunctively specifies the domain of existence**.

Thus, in *A=wa B=de aru*, the constituent *B* denotes a conceptual domain of location, whereas in *A=wa B=ni aru/iru*, *B* denotes a physical domain of location. This difference surfaces formally when B is replaced by a pro-form. As shown in (25) and (26) below, in the former construction *B* is substituted by the pro-form ***sō***, which expresses such notions as property, attribute, characteristic or manner and is glossable as 'so', whereas in the latter construction *B* is substituted by the locative demonstrative pronoun ***soko*** 'there':

(25) *A=wa **B**=de aru.* → *A=wa **sō**=de aru.* (conceptual domain)

(26) *A=wa **B**=ni aru/iru* → *A=wa **soko**=ni aru/iru.* (physical domain)

Although both constructions are assumed to involve a proposition of the form IN(x, y), the formal opposition between *sō* and *soko* reflects whether the second term *y* denotes a conceptual or, alternatively, a physical domain. Note that even when the postposition–conjunction complex *=de* — namely, the eventuality-domain specifier — is used, if *B* in *B=de* denotes a physical domain, it is substituted by *soko*, as in *kare=wa **gakkō**=de / **Tōkyō**=de hataraite=iru* 'he is working in a/the school / in Tokyo' → *kare=wa **soko**=de hataraite=iru* 'he is working there'.

5. The syntactic optionality and pragmatic necessity of *B=de* in *A=wa B=de aru*

Although *B=de* — underlyingly *B=ni-te* — in *A=wa B=de aru* is assumed to be **syntactically**

an adjunct, it **cannot be deleted**. This is because, in this construction, **the most crucial logical** — though not syntactic — **predicate is** not the verbal stem *ar-* that introduces *A*'s existence, but rather **the logically predicate-like postposition *=ni* within the postposition–conjunction complex *=de*, which locates *A*'s existence in the conceptual domain denoted by *B***. It is also because the sentence, while structurally consisting of the obligatory *A=wa aru* 'A exists' and the adjunctive *B=de* 'with (A's existence) located in the conceptual domain of B' or, more practically, 'and it is as B', functions as a categorical or identificational sentence in which **the focus falls on the adjunct *B=de***.

Therefore, the **non-deletability of *B=de*** is assumed to **arise from its pragmatic and informational indispensability** rather than from syntactic necessity. The distinction can be represented as in the following table (27):

(27)

Level	Structure	Function
Syntax	[$_{\text{VP}}$ *A ar-*] + [$_{\text{ConjP}}$ *B=ni-te*]	One-argument VP + adjunctive ConjP
Semantics	'A exists, with A's existence located in the conceptual domain of B'	Existence introduction + domain specification $E(A) \wedge IN(rE(A), B)$
Information Structure	[*A=wa aru*] + [*B=ni-te*]	Given + New (default)
Pragmatics	Focus on *B=ni-te* (default)	Pragmatic obligatoriness and non-deletability of *B=de* due to focus

Japanese has a intransitive verb meaning 'to lead one's life', namely *kuras-u*. In the natural example in (28), if the physical-locative ConjP *Tōkyō=de* '(and it is) in' is deleted, as in (29):

(28) *Kare=wa* ***Tōkyō=de*** *kurasi-te ir-u.* 'He leads his life (and it is) in Tokyo.'

(29) *Kare=wa kurasi-te ir-u.* ' He leads his life.'

the resulting sentence is pragmatically infelicitous. One might therefore be tempted to treat the ConjP *Tōkyō=de* as an argument. If *Tōkyō=de* were an argument, the argument structure of the stative verb *kuras-u* would be [location, theme], where the Theme denote an entity that participates in a state characterisable as a life. However, in (28) the ConjP *Tōkyō=de* can be replaced either by the manner AdvP *siawase-ni* 'happily' or by the comitative PP *kazoku=to* '(lit.) with the family', yielding fully natural results, as in (30) and (31):

(30) *Kare=wa* ***siawase-ni*** *kurasi-te i-ru.* 'He leads his life (and it is) happily.'

(31) *Kare=wa* ***kazoku=to*** *kurasi-te i-ru.* 'He leads his life (and it is) with his family.'

If one were to assum that the physical-locative ConjP *Tōkyō=de* is an argument, then the manner AdvP *siawase-ni* and the comitative PP *kazoku=to* would have to be treated as arguments. But then, what would be the θ-role instead of Location in [location, theme]? The verb *kuras-u* is semantically self-contained in the sense of 'to lead one's life' and does not inherently require any argument for semantic completeness. By default, anyone who is alive is leading a life. Presenting only this content by means of the predicate *kuras-u* is therefore nearly vacuous in terms of information and pragmatics, so the hearer naturally demands some supplementary information such as location, manner, comitative participant, etc. This is why (29) is felt to be infelicitous. Thus, the ConjP *Tōkyō=de* in *Tōkyō=de kuras-u* is best treated as an physical-locative adjunct rather than an argument. The same point applies to the one-argument existential verb *ar-u* 'to exist'. The clause *A=wa aru* 'A exists' is, on its own, informationally and pragmatically nearly vacuous in many cases, unless used in contexts such as *kanōsei=wa aru* 'the possibility exists'. Hence, this language characteristically supplies a domain specification of A's existence, yielding *A=wa B=de aru* 'A exists, with A's existence located within the conceptual domain of B', where the domain of existence is specified by *B=de*.

By contrast, the seemingly near-synonymous verb *sum-u* 'to have one's residence (somewhere)' behaves differently. In the natural example in (32), if the physical-locative PP *Tōkyō=ni* 'in Tokyo' is deleted, as in (33):

(32) *Kare=wa* ***Tōkyō=ni*** *sun-de ir-u.* 'He has his residence in Tokyo.'

(33) *Kare=wa sun-de ir-u.* 'He has his residence.'

the result is pragmatically infelicitous. (Note that the verb *sum-u* is usually translated as 'to live (somewhere)'. Here, the translation 'have one's residence' is adopted here to mirror its argument-selecting profile of *sum-u* in English.) But, unlike *kuras-u* 'to lead one's life', the missing locative PP *Tōkyō=ni* in (33) cannot be substituted by either the manner AdvP *siawase-ni* 'happily' or the comitative PP *kazoku=to* 'with his family' for semantic completeness. The resultant sentences in (34) and (35) remain pragmatically infelicitous:

(34) *Kare=wa* ***siawase-ni*** *sun-de ir-u.* 'He has his residence happily.'

(35) *Kare=wa* ***kazoku=to*** *sun-de ir-u.* 'He has his residence with his family.'

This contrast indicates that, unlike *kuras-u* 'to lead one's life', *sum-u* is not semantically self-contained in the sense of 'to have one's residence (somewhere)'. It thus selects as an argument a physical-locative phrase denoting the place of residence. The argument structure is [location, theme], and the locative is constituted not by the ConjP *NP=de* — underlyingly *NP=ni-te* — but by the PP *NP=ni*.

The adjunct status of the ConjP *Tōkyō=de* in *kare=wa Tōkyō=de kurasi-te iru* in (28), where the verb *kuras-u* does not select any argument but is informationally and pragmatically nearly vacuous on its own, and the argument status of the PP *Tōkyō=ni* in *kare=wa Tōkyō=ni sun-de ir-u*, where *sum-u* selects a locative phrase as an argument, are consistent with the generalisation discussed in Section 2: an argumental *NP=ni* is intrinsically selected by the predicate as semantically indispensable, whereas an adjunct *NP=ni* is extrinsic and contingent to the predicate, hence only optionally linked either to the predicate itself or to some node within the predicate phrase; this optional linkage must be made overt by the adjunctive–conjunction suffix *-te* in Contemporary Japanese. It follows that **simple non-deletability vs. deletability is not, by itself, a reliable diagnostic for argumenthood**.

6. Shift of focus and foregrounding of the lexical meaning 'to exist' of the verb *aru*

In the construction *watasi=wa ryōsikitekina syakaizin=de aru* in (36):

(36)	watasi=wa	ryōsikitekina syakaizin=de	**ar**-u
	I(SUBJ)=TOP	rational member of society=CONCEP-LOC–CONJ(ADJUNCT)	**exist**-PRES
	'I EXIST, with my existence located in the domain of "rational member of society".'		

which can be schematically represented as **E(A) ∧ IN(rE(A), B)**, the existence-introducing component E(A) — that is, the lexical function encoded in the verbal stem *ar-* 'exist' — is backgrounded as given information, whereas the existence-domain-specifying component IN(rE(A), B) is foregrounded as the focus of the utterance. Furthermore, as noted in Section 2, the verb *aru* is more commonly used as a two-argument existential or locative verb than as a one-argument existential verb in contemporary Japanese, and in everyday registers the verb *sonzai-suru* is often preferred for this type of predication. As a result, the meaning 'to exist' is assumed to be rarely brought to the speaker's or hearer's awareness in ordinary categorical or identificational contexts of the type 'A = B' or 'A ∈ B'.

By contrast, in example (37), which takes the form **A WANT [E(A) ∧ IN(rE(A), B)]**, the situation is fairly different:

(37)	watasi=wa	ryōsikitekina syakaizin=de	**ari-**	-tai-T°Ø[3]
	I(SUBJ)=TOP	rational member of society=CONCEP-LOC–CONJ(ADJUNCT)	**exist**	want-PRES
	'I want to EXIST, with my existence located in the domain of "rational member of society".'			

Here, what is at issue is 'one's mode of existence', that is, an orientation toward a particular manner of being. In such a context, **E(A) is no longer backgrounded but instead becomes part of the focus as well as IN (rE(A), B)**, and **the lexical meaning 'to exist' encoded in *ar-* emerges as perceptually salient**. This pragmatic shift of focus, which foregrounds the existential meaning of *ar-*, supports empirically the validity of analysing **the semantic structure of *A=wa B=de aru* not as BE(A, B) but as EXIST(A) ∧ IN(rEXIST(A), B)**.

It should be noted that the English translation *I want to BE a rational member of society*, representable as **A WANT [BECOME(A, B)]** — based on *I AM a rational member of society*, which is representable as **BE(A, B)** — is a complete mistranslation of the Japanese construction *watasi=wa ryōsikitekina syakaizin=de ARI-tai* in (37). This in itself clearly demonstrates that the Japanese *=da* or *=de aru* is not equivalent to the English *be*, nor does it function as a copula in the same way. The difference between them can be represented as in (38):

(38)	*Watasi=wa ryōsikitekina syakaizin=**de aru***:	**EXIST(A) ∧ IN(rEXIST(A), B)**
	*I **am** a rational member of society*:	**BE(A, B)**
	*Watasi=wa ryōsikitekina syakaizin=**de ari-tai***:	**A WANT [EXIST(A) ∧ IN(rEXIST(A), B)]**
	*I **want to be** a rational member of society*:	**A WANT [BECOME(A, B)]**

What corresponds in Japanese to the English *I want to be a rational member of society* in (26) is the sentence in (39):

[3] Tense°Ø denotes the phonologically null allomorph *-0* of the [−past] tense suffix, canonically realised as *-ru*, as noted in Section 10.

(39) *Watasi=wa ryōsikitekina syakaizin=**ni nari-tai***: **A WANT [BECOME(A, B)]**

which can be represented as shown in (40), containing the two-argument state-change verb *naru* 'to become', with *=ni* marking the argument GOAL:

(40)	watasi=wa	ryōsikitekina syakaizin=ni	**nari-**	-tai-T⁰Ø
	I(SUBJ)=TOP	rational member of society=GOAL(ARGUMENT)	**become**	want-PRES
	'I want to BE a rational member of society.'			

7. From Old Japanese to Contemporary Japanese forms

In Old Japanese (OJ, 700–800), the modern formally distinct constructions *A=wa B=**de aru*** 'A exists (and it is) as B' and *A=wa B=**ni aru/iru*** 'A is (located) in B' would have corresponded to the seemingly identical construction *A B=**ni ari***. "Dative *ni* is the general oblique case, marking both argument and non-argument oblique nominals" (Frellesvig 2010: 126), unlike in Contemporary Japanese, where *=ni* marks arguments, while *=de* — which originated from the postposition–suffix sequence *=nite* — marks adjuncts, in accordance with the diachronic description "OJ *ni* and *nite* became more specialized with *ni* being used more for arguments and *nite* more for adjuncts; from mid to late EMJ *nite* acquired the variant *de* which is still in use in contemporary NJ" (*ibid.*: 243), where EMJ and NJ abbreviate Early Middle Japanese (800–1200) and 'new Japanese' — namely, Modern Japanese, 1600– — respectively.

Thus, the former *A B=ni ari* 'A exists (and it is) as B' can be analysed as an existence-introducing and existence-domain-specifying construction consisting of the core construction *A ari* 'A exists' — with the verb *ari* being used as **one-argument existential** predicate — and the conceptual-locative adjunct *B=ni* 'in the conceptual domain referred to by B', as shown in (41). The latter *A B=ni ari* 'A is located in B', on the other hand, can be analysed as a location-specifying-only construction — in which *ari* is employed as a **two-argument locative** predicate — and *B=ni* 'in B' constitutes the physical-locative argument, as indicated in (42):

(41)	A	B=**ni**	**ari**-T⁰Ø
	A(SUBJ)=TOP	B=CONCEP-LOC(**ADJUNCT**)	**exist**-PRES
	'A exists, A's existence located in the conceptual domain of B.'		

(42)	A	B=**ni**	**ari**-T⁰Ø
	A(SUBJ)=TOP	B=PHYSIC-LOC(**ARGUMENT**)	**be located**-PRES
	'A is located in the physical domain of B.'		

The Old Japanese constructions in (41) and (42) appear superficially identical, but when the element *B* is substituted by a pro-form, they diverge as in (43) and (44):

(43) *A* ***B****=ni ari.* → *A* ***sa****=ni ari.* (conceptual domain)

(44) *A* ***B****=ni ari.* → *A* ***soko****=ni ari.* (physical domain)

Consequently, although both constructions is assumed to involve a proposition of the form IN(x, y), the pro-form opposition between *sa* and *soko* reflects whether the second term *y* refers to a conceptual domain or to a physical domain.

These superficially identical constructions in (41) and (42), however, diverged diachronically into the formally distinct modern constructions in (45) and (46):

(45) A=wa B=**ni-te** (surfacing as B=**de**) **ar**-u
A(SUBJ)=TOP B=CONCEP-LOC–CONJ(**ADJUNCT**) **exist**-PRES
'A exists, with A's existence located in the conceptual domain of B.'

(46) A=wa B=**ni** **ar**-u / **i**-ru
A(SUBJ)=TOP B=PHYSIC-LOC(**ARGUMENT**) **be located**-PRES
'A is located in the physical domain of B.'

As already stated in Section 4, in Contemporary Japanese as well, the element *B* undergoes different pro-form substitutions, as in (47) = (25) and (48) = (26):

(47) *A=wa* ***B****=de aru.* → *A=wa* ***sō****=de aru.* (conceptual domain)

(48) *A=wa* ***B****=ni aru/iru* → *A=wa* ***soko****=ni aru/iru.* (physical domain)

In (46), the locative verbs *aru* and *iru*, both semantically corresponding to the Old Japanese *ari* — though the latter originally derived from another verb *wiru* 'to sit; to be seated' — are used for inanimate and animate entities, respectively. The historical development from (33) to (37) is as follows. In Old Japanese, there existed "analytic forms consisting of an infinitive (*ni* …) followed by *ar*-" (Frellesvig 2010: 95) for the so-called copula — that is, *ni ar*-. In the subsequent Early Middle Japanese (EMJ, 800–1200) period, this analytic form "*ni ar*- had the variant *nite ar*-" (*ibid.*: 235). "In the course of EMJ" (*ibid.*: 235), or more specifically "from the end of the period" (*ibid.*: 233), "*nite ar*- became *de ar*-" (*ibid.*: 235). As illustrated in "Table 12.8.

Development of the copula and adjectival-copula paradigms" (*ibid.*: 344), from the end of EMJ to early Late Middle Japanese (early LMJ, Kamakura period 1185-1333), the three forms *ni ar-*, *nite ar-* and *de ar-* coexisted, although "already from late EMJ" (Insei period, 1086-1185), "*ni* was getting more restricted" and "*de* was used far more widely" than *nite*. The same table (*ibid.*: 344) shows that only the newest *de ar-* continued to be used from late LMJ (Muromachi period 1333-1573) into Contemporary Japanese.

The historical developments in (49) and (50):

(49) *A B=**ni** ari* → *A B=**ni-te** ari* → *A=wa B=**de** aru* — E(A) ∧ IN(rE(A), B)

(50) *A B=**ni** ari* → *A B=**ni** ari* → *A=wa B=**ni** aru/iru* — IN(A, B)

may plausibly reflect the diachronic split between *=ni*, used for arguments, and *=ni-te/=de* used for adjuncts, where the conjunctive suffix *-te* adjoins the PP [NP + *=ni*] as an adjunct, rather than as an argument, to another node, as discussed in Section 2. This suggests a shift from a stage in which *=ni* marked both arguments and adjuncts, without the help of the conjunction *-te*, to one in which *=ni* and *=ni-te/=de* became specialised as argumental and adjunctive markers, respectively. Thus, the historical split of the Old Japanese *=ni ari* between the Modern Japanese *=de aru* and *=ni aru/iru* may not reflect a reanalysis of a single construction, but rather the formal divergence of two distinct semantic structures — 'existence and its location in a conceptual domain' vs. 'location in a physical domain' — that happened to share the same surface form *A B=ni ari* in Old Japanese — a possibility that merits further investigation.

8. Default partial negation and the scope of negation marked by the particle *=wa*

The **default negation pattern of the existence-introducing and existence-domain-specifying** sentence in (51) = (18):

(51) kare=wa gakusei=de ar-u
he(SUBJ)=TOP student=CONCEP-LOC–CONJ(ADJUNCT) exist-PRES
'He exists, and it is as a student.'

which, in line with Section 2, can be schematically represented as the conjunctive proposition E(A) ∧ IN(rE(A), B), is the **partial negation** in (52):

(52)	kare=wa	gakusei=de=**wa**	(*ara-)	**-nai**-T^{o}Ø
	he(SUBJ)=TOP	student=CONCEP-LOC–CONJ(ADJUNCT)=**SCOPE OF NEG**	exist	**NEG**-PRES
	'He exists, but it is **not** as a student.'			

rather than the total negation in (53):

(53)	kare=wa	gakusei=de	(*ara-)	**-nai**-T^{o}Ø
	he(SUBJ)=TOP	student=CONCEP-LOC–CONJ(ADJUNCT)	exist	**NEG**-PRES
	'He does **not** exists as a student.'			

Note that in (52) and (53), the stem of the existential verb *aru* is represented as (*ara-), since in contemporary Standard Japanese the negative form of this verb is not the theoretically expected **ara-nai*, but obligatorily the suppletive adjective *nai*. The more natural sentence *kare=wa gakusei=de=**wa** **nai*** in (52) — or, more colloquially, *kare=wa gakusei=dy**a** **nai***, with the contracted form *=dya* derived from the sequence *=de=wa* — does **not negate the existence of *kare* itself**, but rather **negates only *gakusei=de* as the relevant domain of existence**. In this way, **the particle *=wa* in Japanese can mark the scope of negation**.

Schematically, the partial negation *A=wa B=de=**wa** **nai*** in (52) — underlyingly, **A=wa B=de=**wa** ara-**nai*** — can be represented as **E(A) ∧ ¬IN(rE(A), B)**, whereas the total negation *A=wa B=de **nai*** in (53) — underlyingly, **A=wa B=de ara-**nai*** — can be represented as **¬(E(A) ∧ IN(rE(A), B)) ≡ ¬E(A) ∨ ¬IN(rE(A), B)**. Although this disjunction is compatible with ¬E(A) ∧ IN(rE(A), B), E(A) ∧ ¬IN(rE(A), B) or ¬E(A) ∧ ¬IN(rE(A), B), the first case is inconsistent with the conditional IN(rE(A), B) ⇒ E(A) — hence, the contrapositive ¬E(A) ⇒ ¬IN(rE(A), B) — and the third case, ¬E(A) ∧ ¬IN(rE(A), B) ≡ ¬E(A), is pragmatically infelicitous in topicalised constructions with *A=wa*, since *A=wa*, as a topicalised phrase, tends to presuppose the contextual or situational givenness — that is, the existence — of *A*, and since there are dedicated Japanese forms for existential negation such as *A=wa nai / i-nai / sonzai-si-nai*; in practice, therefore, the interpretation converges on E(A) ∧ ¬IN(rE(A), B). Hence, the former pattern in (52), which unambiguously denotes the partial negation E(A) ∧ ¬IN(rE(A), B), is judged to be the more natural one.

This analysis is further supported by Kansai variety forms such as *gakusei=ya nai* and *gakusei=ya ara-hen*, which are natural and frequent and correspond closely to the standard forms *gakusei=de=wa nai* and *gakusei=dya nai*. In the former forms, the focus particle *=ya* originated from the sequence *=de=wa*, via *=dya*, and *-hen* functions as a negator, just as *-nai* does. By contrast, corresponding forms without this particle — namely, *gakusei=de nai* and *gakusei=de ara-hen* — are pragmatically infelicitous.

If *=de aru* were a single copula, then *A=wa B=de aru* would denote BE(A, B), and its negation would take the form of plain predicate negation — namely, *A=wa B=de nai*, representable as ¬BE(A, B), with no *=wa* following *=de*. **Empirically, however, *A=wa B=de nai* is perceived as infelicitous, whereas *A=wa B=de=wa nai* is preferred**. This spontaneous preference for the particle *=wa* after *=de* is unexpected under the simple-copula analysis, but follows naturally if the sentence is decomposed as E(A) ∧ IN(rE(A), B), where the logically and pragmatically most plausible interpretation is the partial negation E(A) ∧ ¬IN(rE(A), B).

By contrast, the **default negation of the location-specifying-only** (54) = (22) and (55) = (23):

(54)	sore=wa	gakkō=ni	ar-u
	it(SUBJ)=TOP	school=PHYSIC-LOC(ARGUMENT)	be located-PRES
	'It is located in the school.'		

(55)	kare=wa	gakkō=ni	i-ru
	he(SUBJ)=TOP	school= PHYSIC-LOC(ARGUMENT)	be located-PRES
	'He is located in the school.'		

which can both be represented as IN(A, B), is the **total negation** indicated in (56) and (57):

(56)	sore=wa	gakkō=ni	(*ara-)	**nai**-T^{0}Ø
	it(SUBJ)=TOP	school=PHYSIC-LOC(ARGUMENT)	be located	**NEG**-PRES
	'It is **not** located in the school.'			

(57)	kare=wa	gakkō=ni	i-	**nai**-T^{0}Ø
	he(SUBJ)=TOP	school= PHYSIC-LOC(ARGUMENT)	be located	**NEG**-PRES
	'He is **not** located in the school.'			

Unlike the existence-introducing and existence-domain-specifying conjunctive E(A) ∧ IN(rE(A), B), the location-specifying-only IN(A, B) is an atomic proposition and, naturally, its negation constitutes ¬IN(A, B), which corresponds to (56) and (57).

On the other hand, when the PP *gakkō=ni* in (56) and (57) is marked by *=wa*, this particle functions contrastively, as in (58) and (59), respectively, where CONTR abbreviates contrastive:

(58)	sore=wa	gakkō=ni=**wa**	(*ara-)	**nai**-T^{0}Ø
	it(SUBJ)=TOP	school=PHYSIC-LOC(ARGUMENT)=**CONTR**	be located	**NEG**-PRES
	'It is **not** located in the school **but somewhere else**.'			

(59)	kare=wa	gakkō=ni=**wa**	i-	**nai**-T^{0}Ø
	he(SUBJ)=TOP	school= PHYSIC-LOC(ARGUMENT)=**CONTR**	be located	**NEG**-PRES

'He is **not** located in the school **but somewhere else**.'

In contrast to (56) and (57), which are representable as ¬IN(A, B), (58) and (59) can be represented schematically as ¬IN(A, B) ∧ ∃x (IN(A, x) ∧ x ≠ B) — semantically, 'A is not located in B, and there exists a physical domain x distinct from B such that A is located in x'.

The difference in default negation pattern between the existence-introducing and existence-domain-specifying construction and the location-specifying-only one lies in the fact that the former, **E(A) ∧ IN(rE(A), B), is conjunctive**, whereas the latter, **IN(A, B), is atomic**. This ultimately derives from **the status of B as an adjunct in the former and as an argument in the latter**, since if B were an argument, E(A) ∧ IN(rE(A), B) would correspond to something like EXIST-MANNER(A, B), whose negation would be the total ¬EM(A, B).

Finally, if *=de aru* were a single copular unit, partial negation with *=wa* would necessarily take the form **kare=wa* <u>*gakusei=**wa***</u> *de **nai***, as in (60):

(60)	*kare=wa	gakusei=**wa**	de	**nai**-T^{0}Ø
	he(SUBJ)=TOP	student(PRED)=**CONTR**	COP	**NEG**-PRES

'He is **not** a student **but something else**.'

This is ungrammaticality further supports the separability of *=de* and *aru* and the constituency of *gakusei=de*.

9. The differences between *A=wa B=de aru* and *A=ga B=de aru*

The following table (61):

(61)

Sentence	Propositional structure	Informational structure	
		Given	New
*A=**wa** B=de aru*	E(A) ∧ IN(rE(A), B)	E(A)	IN(rE(A), B)
*A=**wa** B=de=wa nai*	E(A) ∧ ¬IN(rE(A), B)	E(A)	¬IN(rE(A), B)
*A=**wa** B=de=wa naku C=**wa** B=de aru*	E(A) ∧ ¬IN(rE(A), B) ∧ E(C) ∧ IN(rE(C), B)	E(A) ∧ E(C)	¬IN(rE(A), B) ∧ IN(rE(C), B)
*A=**ga** B=de aru*	E(A) ∧ IN(rE(A), B)	∃x (E(x) ∧ IN(rE(x), B))	x = A
*A=**ga** B=de=wa nai*	E(A) ∧ ¬IN(rE(A), B)	∃x (E(x) ∧ IN(rE(x), B))	x ≠ A
*A=**ga** B=de=wa naku C=**ga** B=de aru*	E(A) ∧ ¬IN(rE(A), B) ∧ E(C) ∧ IN(rE(C), B)	∃x (E(x) ∧ IN(rE(x), B))	(x ≠ A) ∧ (x = C)

provides a preliminary overview of the propositional identity and the information-structural differences between sentences of the type ***A=wa B=de aru*** and those of the type ***A=ga B=de aru***, without attempting a full formalisation, which lies beyond the scope of this paper.

Syntactically, the structure of *A=**ga** B=de aru* in (61) can be represented as in (62):

(62) $[_{TP}\ \boxed{[_{NP}\ A_i{=}ga]}\ [_{T'}\ \boxed{[_{VP_1}\ [_{ConjP}\ B{=}ni\text{-}te]\ [_{VP_0}\ [_{NP}\ x_i]\ [_{V}\ ar\text{-}]]]}\ [_{T}\ \text{-}ru]]]$

where *x* represents a trace, used here instead of the conventional *t*. In this structure, the higher VP $[_{VP_1}\ [_{ConjP}\ B{=}ni\text{-}te]\ [_{VP_0}\ [_{NP}\ x_i]\ [_{V}\ ar\text{-}]]]$ denotes the given information ∃x (E(x) ⋀ IN(rE(x), B)), whereas the **untopicalised** — and thus capable of indicating a piece of new information — NP $[_{NP}\ A_i{=}ga]$ denotes the new information x = A, through the anaphoric relation of the trace $[_{NP}\ x_i]$ to its antecedent $[_{NP}\ A{=}ga_i]$. The constituents representing given and new information are enclosed within boxes and, furthermore, those representing new information are shaded. (The bracket notation adopted here is a descriptive representation of hierarchical constituency, and the traditional category labels are used merely as a device for visualising the structure.)

On the other hand, the structure of *A=**wa** B=de aru* in (61) can be represented as in (63):

(63) $[_{TopP}\ \boxed{[A_i{=}wa]}\ [_{Top'}\ [_{TP}\ [_{VP_1}\ \boxed{[_{ConjP}\ B{=}ni\text{-}te]}\ \boxed{[_{VP_0}\ [_{NP}\ x_i]\ [_{V}\ ar\text{-}]]}]\ [_{T}\ \text{-}ru]]\ [_{Top}\ Top^{0}\emptyset]]]$

The category *Topic*, abbreviated as *Top* in (63), is "a kind of functional category" that "can be considered to embed the IP into discourse and context" (Hasegawa 1999: 91, translated from Japanese), and is assumed in this paper to be realised by the phonologically null morpheme *-0*, which we denote as Topic^{0}Ø, abbreviated as Top^{0}Ø. The element that moves to the specifier of TopP is assigned the particle *=wa* and functions to indicate the topic of the sentence. In this structure, the **topicalised** phrase $[A_i{=}wa]$ — which typically indicates a piece of given information — and the lower VP $[_{VP_0}\ [_{NP}\ x_i]\ [_{V}\ ar\text{-}]]$ — where the verb $[_{V}\ ar\text{-}]$ 'to exist' provides another piece of given information, since the topicalised phrase $[A_i{=}wa]$ generally presupposes the existence of A — together denote the given information E(A). By contrast, the ConjP $[_{ConjP}\ B{=}ni\text{-}te]$ denotes the new information ⋀ IN(rE(A), B), since a sentence conveying only given information would be informationally vacuous.

Note that, since IN(rE(x), B) ⇒ E(x), it follows that ∃x (E(x) ∧ IN(rE(x), B)) ≡ ∃x IN(rE(x), B); nevertheless, the logically redundant ∃x (E(x) ∧ IN(rE(x), B)) represents a component within the informational structure that corresponds to the syntactic–semantic structure assumed in this paper.

10. The contracted form =*da*

In spoken language, the contracted form =*da* is very often used instead of =*de aru*, which is more formal and literal, as in *kare=wa gakusei***=*da*** rather than *kare=wa gakusei***=*de aru***. It should be noted that this **contracted form =*da*** does not represent a new copular category 'be' but is a **purely phonological reduction of the postpositional–conjunctive–verbal complex =*de aru***, as shown in (64):

(64)	kare=wa	gakusei=**d**	**a**-T⁰Ø-Top⁰Ø
	he(SUBJ)=TOP	student=**CONCEP-LOC–CONJ**(ADJUNCT)	**exist**-PRES
	'He exists, with his existence located in the conceptual domain of "student".'		

The historical change from =*de aru* to =*da*, possibly via =*dea*[4], involves no semantic or syntactic reanalysis, but a process of phonological reduction comparable to German *in dem* → *im* 'in the' or Portuguese *de a* → *da* 'from the' or 'of the'. Just as there is no semantic necessity for *in* and *dem* to fuse into *im*, or for *de* and *a* into *da* — a phenomenon comparable to the English *in the*, *from the* or *of the* forming a single word — there is likewise no semantic motivation for =*de* and *aru* to form a SINGLE lexical item.

That said, however, the reduction of *aru* to *a* can be accounted for in pragmatic and informational terms. First, as already dicussed in Sections 2 and 5, the crucial logical — though not syntactic — predicate in this construction is not the verbal predicate *ar*-, representable as E( ), but the underlying predicate-like postposition =*ni*, representable as IN(r( ),), contained within the surface postposition–conjunctive complex =*de*.

Second, the topicalised phrase *A*=*wa*, marked by the topicalisation particle =*wa*, generally presupposes at the discourse level the existence of *A*, namely E(A). Moreover, the adjunctive-conjunction phrase *B*=*de* is representable as the additional conjunct ∧ IN(r( ), B). When *A*=*wa* and *B*=*de* occur in succession in the string *A*=*wa B*=*de*…, the hearer is led to anticipate the full

[4] Historically, the form =*dea*, which yielded the modern =*da*, can be attested in Azuchi–Momoyama-period materials: *bonnin yorimo giŭzaini fuxôzuru coto***dea** "(it) is a reason for imposing a heavier guilt (on him) than on an ordinary person" (Amakusa Edition of *Esopo no fabulas*, 1593, p.475, ll.10-11, in Text Data Sets for Research on the History of Japanese).

conjunctive proposition E(A) ∧ IN(rE(A), B). In this context, the existential verb *aru* becomes informationally redundant when it once again introduces E(A).

In non-sentence-final coordinate clauses, the inflected form *ari* of the verb *aru* in *=de ari* is often omitted, which is a natural consequence of the above anticipation. Returning to Section 1, the compound sentence *kare=wa gakusei=de, kanozyo=wa syakaizin=d**a*** 'he (is) a student, and she is a working adult' in (5) — underlyingly, *kare=wa gakusei=de **ari**, kanozyo=wa syakaizin=de **aru*** — can be analysed as in (65):

(65)	kare=wa	gakusei=de	(**ari**)	kanozyo=wa	syakaizin=d	**a**-T^{0}Ø-Top^{0}Ø
	he(SUBJ)=TOP	student=LOC-CONJ	(**exist**)	she(SUBJ)=TOP	working adult= LOC-CONJ	**exist**-PRES

'He (**exists**) with his existence located in the conceptual domain of "student",
she **exists** with her existence located in the conceptual domain of "working adult".'

The construction in (65) is an instance of zeugma, in which the inflected form *ari* of the existential verb aru is deleted, since, as stated in the previous paragraph, the proposition E(kare) ∧ IN(rE(kare), gakusei) is predictable already at the point *kare=wa gakusei=de*, and since *ari* in *gakusei=de ari* is contextually recoverable on the basis of the structural parallelism between the two coordinate clauses. Therefore, *=de* is not an inflected form of *=da*.

Third, in a sentence in which IN(rE(A), B) is presented as new information, E(A) is presupposed and thus treated as given, owing to the conditional IN(rE(A), B) ⇒ E(A). In this context, E(A) is pragmatically and informationally less significant than the focus component IN(rE(A), B).

It may therefore be assumed that **this low pragmatic and informational salience is reflected in the phonological reduction of *aru* to *a***. This reduction can be analysed as follows: morphologically, the [−past] tense suffix has the phonologically null allomorph /-0/ — which we denote as Tense^{0}Ø, abbreviated as T^{0}Ø — instead of the canonical -*u* /-ru/ in *ar-u* /ar-ru/, yielding /ar-0/; phonologically, the word-final /-r/ is deleted in phonetic implementation, giving ris+e to the surface form *a*.

On the other hand, The reduction of the underlying *=ni-te ar-0* to the surface *=da*, via *=de ar-0*, can be accounted for by the fixed order of these concatenated morphemes and its frequency. In existence-introducing and existence-domain-specifying constructions, the morphemes *=ni*, *-te* and *ar-* — which as a whole corresponds to the schematic expression E( ) ∧ IN(rE( ),) and in which *=ni*, *-te* and *ar-* represent IN(r( ),), ∧ and E( ), respectively — is invariably ordered in this way, except in scrambling constructions — or *kakimazekōbun* in Japanese linguistics — such as *B***=de** *watasi / watasitati / wareware=wa* ***ari****-tai* '(lit.) as B, I / we / we want to exist'. **The very high frequency of this fixed order in actual utterances is assumed to have led to the phonological reduction of *=ni-te ar-u* → *=de ar-u* → *=de-a* → *=d-a***, just as in the German and Portuguese examples mentioned above, and likewise in the contractions *si-te=i-ru* → *si-te-ru* 'to be doing something' or 'to have done sth.', and *si-te=sima-u* → *si-t-sima-u* → *si-t-sya-u* 'to end up doing sth.' or 'to do sth. inadvertently or unintentionally' in colloquial Japanese.

It is not uncommon for a phonological form whose linear segmentation is no longer transparent to encode multiple meanings. In French, for instance, the present indicative first-person singular form *ai* /e/ of the verb *avoir* 'to have', as in *j'ai* 'I have' — derived from the same inflected form *habeō* /hab-ē-ō/, plausibly surfacing as [habeoː], from the Classical Latin *habēre* '*idem.*' — expresses the lexical meaning 'have' in addition to a bundle of functional meanings such as [−non-assertive], [−past], [−posterior], [−anterior], [−non-participant], [−addressee] and [−plural]. Japanese *=da*, too, is nothing more than the outcome of diachronic phonological attrition: four independent morphemes *=ni*, *-te*, *ar-* and *-ru* have been worn down over time and have come to assume a simple shape *=da*.

In sum, ***=da*** as a whole should not be regarded as a copula in the strict sense, but rather as a **phonologically reduced surface realisation of the underlying structure *=ni-te ar-0***. In this underlying structure, *=de* — underlyingly *=ni-te* — constitutes an adjunctive postposition–conjunction complex that conjunctively specifies the conceptual domain in which the theme's existence is located, while *ar-* functions as the one-argument existential predicate and *-0* realises the phonologically null [−past] tense suffix $T^{o}Ø$. This is precisely why the following highly colloquial dialogue in (66) is possible:

(66) *Kare=tte gakusei=**da** yone? — Iya, gakusei=tte yū=ka, gakusei=**de**=mo **aru**=tte kanzi kana.*

'He's a student, right?' — 'Well, if you can call him a student, I'd say he's *also* a student, kind of.'

In (57), the interlocutor, recognising that the underlying structure of *gakusei=da* is *gakusei=de aru*, when asked to confirm that *kare*'s domain of existence can be identified with that of 'student', replies that it extends to other domains as well, expressed by the form *gakusei=de=**mo** aru*, representable as E(he) ∧ IN(rE(he), student) ∧ IN(rE(he), something else)….

11. Syntactic and logical-propositional differences between NPs with *=da*, nominal adjectives in *-da* and verbal adjectives

Japanese **nominal adjectives** (**NAs**), also referred to as ***na*-adjectives** due to their attributive forms ending in *-na* — the so-called *rentaikei* in Japanese linguistics — as in *yukai-**na** hito* 'a pleasant person', have often been analysed as consisting of "so-called NA stems plus the so-called copula *=da*", in parallel with "NPs plus the so-called copula *=da*", as in *sore=wa **yukai=da*** 'it is pleasant', in parallel with *sore=wa **gakkō=da*** 'it is a school'.

According to Makino (2025²: 22), however, **the core of NAs lies in the NA stem in *-ni***, as in *yukai-ni* 'to be pleasant' and *suki-ni* 'to like'. The element *-ni* is not a postposition but the NA thematic suffix, which functions as an indispensable morphological and semantic constituent in the formation of the NA stem. These NA stems are lexical predicates that require arguments — the argument structure [theme] inherent in *yukai-ni* and [experiencer, theme] in *suki-ni* — just as verbal-adjective (VA) stems are, such as the one-argument predicate *tanosi-ku* 'to be enjoyable' with [theme] and the two-argument predicate *hosi-ku* 'to want' with [experiencer, theme]. However, outside small clauses functioning as complements of cognition or perception verbs, as in *watasi=wa* [*sore=o yukai-**ni***] *omot-ta / kanzi-ta* 'I found / felt [it pleasant]', such NA stems cannot function as predicates in their bare *-ni* form unlike VA stems, as in *sore=wa tanosi-**ku**, *yukai-**ni*** 'it is enjoyable, pleasant' and *sore=wa *yukai-**ni**, tanosi-**i*** 'it is pleasant, enjoyable'.

Consequently, an NA stem forms a nominal-adjective phrase accompanied by PRO, as in [*PRO yukai-ni*], which serves as the complement of the adjunctive-conjunction *-te*, yielding a ConjP [*PRO yukai-ni-te*]. This ConjP is then adjoined to the VP headed by the existential-verb stem *ar-* within a one-argument obligatory construction, as in [*sore*$_i$*=wa t*$_i$ *ar-0-0*] 'it exists', where *-0-0* represents the sequence of T⁰Ø and Top⁰Ø. Only in this configuration can the NA stem function as the predicate of an independent sentence, as in [*sore*$_i$*=wa* [*PRO*$_i$ *yukai-ni-te*] *t*$_i$ *ar-0-0*], which surfaces as *sore=wa yukai-d-a* [*sore*$_i$*=wa* [*PRO*$_i$ *yukai-d*] *t*$_i$ *a-0-0*] "(lit.) it, being pleasant, exists'. On the other hand, the so-called copular construction *sore=wa gakkō=d-a* '(lit.) it exists, with its existence located in the conceptual domain of "school"' can be underlyingly analysed, as already discussed, as [*sore*$_i$*=wa* [*gakkō=ni-te*] *t*$_i$ *ar-0-0*], which surfaces [*sore*$_i$*=wa* [*gakkō=d*] *t*$_i$ *a-0-0*].

Thus, the so-called copular sentences and the nominal-adjectival sentences share the syntactic property that they both consist of an existence-introducing, one-argument core construction combined with either an existence-domain-specifying adjunctive ConjP or a state-specifying adjunctive ConjP, and can both be schematically represented as conjunctive propositions, as in (67) and (68):

(67) *Sore=wa gakkō=d-a.* [*sore*$_i$*=wa* [*gakkō=ni-te*] t_i *ar-0-0*] E(sore) ∧ IN(rE(sore), gakkō)

(68) *Sore=wa yukai-d-a.* [*sore*$_i$*=wa* [*PRO*$_i$ *yukai-ni-te*] t_i *ar-0-0*] E(sore) ∧ PLEASANT(sore)

The difference is that, since the PP [*gakkō=ni*] in (67) does not constitute a lexical predicate, it does not require any argument; therefore, the adjunctive construction [*gakkō=ni-te*] does not contain PRO, unlike the adjunct [*PRO yukai-ni-te*] in the nominal-adjectival construction in (68).

By contrast, **verbal adjectives (VAs)** — also referred to as adjectives, canonical adjectives or ***i*-adjectives** due to their attributive forms ending in *-i*, as in *tanosi-**i** hito* 'a fun person' — form constructions in which they themselves function as the main predicate. Such constructions, as in *sore=wa tanosi-i* 'it is enjoyable', consist only of a state-specifying obligatory construction without any adjunct and can be schematically represented as atomic propositions, as in (69):

(69) *Sore=wa tanosi-i.* [*sore*$_i$*=wa* t_i *tanosi-i-0-0*] ENJOYABLE(sore)

The syntactic and logical-propositional differences between (67) and (68), on the one hand, and (69), on the other, become manifest in their default negation patterns, as shown in (70) – (72):

(70) *Sore=wa <u>gakkō=de**=wa**</u> **na-i**.* E(sore) ∧ ¬IN(rE(sore), gakkō)

(71) *Sore=wa <u>yukai-de**=wa**</u> **na-i**.* E(sore) ∧ ¬PLEASANT(sore)

(72) *Sore=wa <u>tanosi-ku</u> **na-i**.* ¬ENJOYABLE(sore)

The default negation patterns of the constructions in (67) and (68) — representable as the conjunctive propositions E(sore) ∧ IN(rE(sore), gakkō) and E(sore) ∧ PLEASANT(sore) — are partial, namely E(sore) ∧ ¬IN(rE(sore), gakkō) and E(sore) ∧ ¬PLEASANT(sore), as in (70) and (71), whereas that of the construction in (69) — representable as the atomic proposition ENJOYABLE(sore) — is total, namely ¬ENJOYABLE(sore), as in (72).

It follows that the stem allomorph *at-* of the verb *ar-u* occurring in the past form *tanosi-k-**at**-ta*, for example, does not instantiate the one-argument existential verb *ar-* 'to exist' used in both so-called copular and nominal-adjectival sentences. Rather, as stated in Makino (2025[1]: 4), it is an allomorph of "the stem *ar-* from the light verb *ar-u*" inserted as a matter of formal necessity in order to concatenate the **verbal** past-tense suffix *-ta* with the **non-verbal** VA stem *tanosi-ku*, which is incompatible with the former, as in *tanosi-ku* + *-ta* → *tanosi-ku* + ***ar-*** + *-ta* → *tanosi-k-**at**-ta*. The default negation of this past form *tanosi-k-at-ta* is likewise total, namely *tanosi-ku **na-k**-a-ta*, rather than the partial negation *tanosi-ku**=wa** **na-k**-at-ta*. This demonstrates that *tanosi-ku* does not constitute an adjunct forming a separate proposition and thus provides further evidence that *ar-* in this construction is not a one-argument existential verb. Note finally that the partial negation of the construction in (73) gives rise to an implication such as ∧ SOME-OTHER-STATE(sore):

(73) *Sore=wa* <u>*tanosi-ku**=wa***</u> ***na-i***. ¬ENJOYABLE(sore) ∧ SOME-OTHER-STATE(sore)

The syntactic derivation of the so-called copular, nominal-adjectival and verbal-adjectival constructions in (74) = (67), (75) = (68), (76) = (69), and (77) can be represented as follows:

(74) *Sore=wa gakkō=d-a.* '(lit.) It exists, with its existence located in the domain of "school".'

Adjunct	**Core construction**
[*=ni*]	[*ar-*]
→ [***gakkō***=*ni*]	→ [***sore*** *ar-*]
→ [[*gakkō=ni*] ***-te***]	
	→ [**[[*gakkō=ni*] *-te*]** [*sore ar-*]]
	→ [[[[*gakkō=ni*] *-te*] [*sore ar-*]] ***T^{0}Ø***]
	→ [[[[[*gakkō=ni*] *-te*] [*sore ar-*]] *T^{0}Ø*] ***Top^{0}Ø***]
	→ [***sore*$_i$=wa** [[[[[*gakkō=ni*] *-te*] [***t*$_i$** *ar-*]] *T^{0}Ø*] *Top^{0}Ø*]]
	→ *sore=wa gakkō=d-a*

As already noted in the previous section, the underlying sequences *=ni-te-ar-0* and *-ni-te-ar-0* surfaces as *=d-a* and *-d-a* in (74), as well as in (75) below:

(75) *Sore=wa yukai-d-a.* '(lit.) It, being pleasant exists.'

Adjunct	**Core construction**
[*yukai-ni*]	[*ar-*]
→ [***PRO*** *yukai-ni*]	→ [***sore*** *ar-*]
→ [[*PRO* *yukai-ni*] ***-te***]	
	→ [**[[*PRO*$_i$ *yukai-ni*] *-te*]** [*sore*$_i$ *ar-*]]
	→ [[[[*PRO*$_i$ *yukai-ni*] *-te*] [*sore*$_i$ *ar-*]] ***T***^{0}***Ø***]
	→ [[[[[*PRO*$_i$ *yukai-ni*] -te] [*sore*$_i$ *ar-*]] *T*^{0}*Ø*] ***Top***^{0}***Ø***]
	→ [***sore***$_i$***=wa*** [[[[[*PRO*$_i$ *yukai-ni*] *-te*] [***t***$_i$ *ar-*]] *T*^{0}*Ø*] *Top*^{0}*Ø*]]
	→ *sore=wa yukai-d-a*

(76) *Sore=wa tanosi-i.* 'It is enjoyable.'

Core construction

[*tanosi-ku*]

→ [***sore*** *tanosi-ku*]

→ [[*sore tanosi-ku*] ***T***^{0}***Ø***]

→ [[[*sore tanosi-ku*] *T*^{0}*Ø*] ***Top***^{0}***Ø***]

→ [***sore***$_i$***=wa*** [[[***t***$_i$ *tanosi-ku*] *T*^{0}*Ø*] *Top*^{0}*Ø*]]

→ *sore=wa tanosi-i*

(77) *Sore=wa tanosi-k-at-ta.* 'It was enjoyable.'

Core construction

[*tanosi-ku*]

→ [***sore*** *tanosi-ku*]

→ [[*sore tanosi-ku*] ***-ta***]

→ [[*sore tanosi-ku*] ***ar-*** *-ta*]

→ [[[*sore tanosi-ku*] *ar- -ta*] ***Top***^{0}***Ø***]

→ [***sore***$_i$***=wa*** [[[***t***$_i$ *tanosi-ku*] *ar- -ta*] *Top*^{0}*Ø*]]

→ *sore=wa tanosi-k-at-ta*

In the VA non-past form in (76), the sequence [-*ku* + *T⁰Ø*] is morphophonologically realised as -*i*, yielding the surface form *tanosi-i* from the underlying elsewhere form *tanosi-ku*. The exponent -*i* may therefore be analysed either as a contextual allomorph -*i* of the VA thematic suffix -*ku* in the non-past environment or, alternatively, as an amalgamated realisation of the sequence [-*ku* + *T⁰Ø*], as stated in Makino (2025[1]: 14).

In the analysis of the VA past form in (77), the choice between the non-past T⁰Ø, realised as -*0*, and the past suffix -*ta* is treated as a paradigmatic selection made at a single derivational stage, since the [−past] -*0* and the [+past] -*ta* constitute a paradigm whose terms acquire linguistic value solely through their opposition at the same point of speech. When the past suffix -*ta* is selected, it proves incompatible with non-verbal predicates such as *tanosi-ku*, thereby triggering the insertion of the light verb stem *ar*- as a repair strategy. Then, the stem-final vowel -*u* of the suffix -*ku* is deleted before *ar*-, together with the assimilation of *ar*- to *at*- before -*ta*.

12. Classical Latin *esse* and Japanese =*da* — an exploratory note

Japanese is not the only language that employs a one-argument existential verb in a copula-like way. In Classical Latin, like Japanese, the default one-argument existential verb was employed to denote relations between two entities. In Latin, the verb *esse* 'to exist; to be' was used both as a one-argument/two-argument existential verb and as a so-called copula, as exemplified in *Deus est* 'God exists' and *Deus lūx est* 'God is light'. In the latter construction, the NP *lūx* 'light' can be analysed as a predicative apposition to the subject NP *Deus* 'God', marked by the same nominative case. In its primitive sense, this construction can thus be interpreted as 'God, being light, exists' or 'God exists as light', representable as EXIST(Deus) ∧ PRED(Deus, lūx), where PRED abbreviates 'predication' and will be abbreviated as P below. If the initial conjunct E(Deus) is deleted, the remaining proposition P(Deus, lūx) constitutes what is traditionally described as a copula.

In Classical Latin, however, the E(A) component in the conjunctive proposition E(A) ∧ P(A, B) can be taken to have remained present, even if it was typically backgrounded as in the Japanese E(A) ∧ IN(rE(A), B), since, to represent 'to want to be', the state-change verb *fierī* 'to become' was used instead of *esse*, as in:

(78) *A B **fierī** cupit/vult.* 'A wants to **be** B.': A WANT [BECOME(A, B)]

(79) *A B **esse** cupit/vult.* 'A wants to **exist as** B.': A WANT [E(A) ∧ P(A, B)]

The relation between (78) and (79) is closely paralleled by the following opposition between (80) and (81) in Japanese:

(80) *A=wa B=ni **nari**-tai.* 'A wants to **be** B.': A WANT [BECOME(A, B)]

(81) *A=wa B=de **ari**-tai.* 'A wants to **exist as** B.': A WANT [E(A) ∧ IN(rE(A), B)]

In Classical Latin, the tendency to use *fierī* 'to become' rather than *esse* to express 'to want to be' can be compatible with the view that, even in its copula-like use, the component E(A) still remained present.

Thus, Japanese and Classical Latin share the commonalities in (82) and (83):

(82) They employ existing linguistic resources to denote relations between two entities.

(83) One of those resources is the default one-argument existential verb — namely, *ar-u* in Japanese and *esse* in Classical Latin.

On the other hand, they differ in (84) and (85):

(84) The other resources differ — namely, the conceptual-locative postposition *=ni* and the adjunctive-conjunction suffix *-te* in Japanese, and predicative apposition and agreement in Classical Latin.

(85) Whereas, in Classical Latin, the component E(A) could, in principle, be deleted from E(A) ∧ P(A, B) without syntactic difficulty, thereby yielding a genuine copula P(A, B) — even though, in practice, E(A) appears not to have been deleted — in Japanese the component E(A) cannot, as a matter of syntactic structure, be deleted from E(A) ∧ IN(rE(A), B).

Later, the deletion of the E(A) component from the verb *esse* representable as E(A) ∧ P(A, B) may have given rise to a genuine copular use representable as P(A, B). In parallel, the existential use representable as E(A) may have receded, with other expressions for E(A) subsequently developing.

13. Conclusion

In summary, **Japanese has no genuine copula — *=da* is not *be***. Instead, Japanese is **a language that has employed existence, conjunction and location to denote relations between two entities compositionally**. The function of *=da* and its non-contracted form *=de aru* is to indicate that something **exists** (***ar-***) and that this existence is specified **conjunctively** (***-te***) as **located in a conceptual domain** (***=ni***) defined by a category. Japanese linguistically encodes E(A) ∧ IN(rE(A), B) through *=ni*, *-te* and *ar-*, while any interpretation in terms of A = B or A ∈ B arises only as a derived meta-level inference. Thus, treating *=da* as *be* may be practical for pedagogical, translational and surface-level purposes, but **the Japanese grammatical system contains no primitive corresponding to *be***.
Syntactically and semantically, the construction *A=wa B=de aru* consists of an existence-introducing, one-argument obligatory construction [*A=wa aru*] 'A exists' combined with an existence-domain-specifying adjunctive construction [*B=de*] — underlyingly [*B=ni-te*] — meaning 'with (A's existence) located in the conceptual domain of B' or, more practically, 'and it is as B'. In this construction, the most crucial logical — though not syntactic — predicate is not the verbal stem *ar-* that introduces *A*'s existence, but rather the predicate-like conceptual-locative postposition *=ni* within the postposition–conjunction complex *=de*, which locates *A*'s existence in the conceptual domain of *B*.

Consequently, treating *=da*, or its non-contracted form *=de aru*, as a single copula is structurally problematic. The form *=de aru* is more appropriately analysed as an **existence-and-domain complex**, consisting of the one-argument existential verb *aru* 'to exist' and the predicate-like conceptual-locative postposition *=ni* 'in' plus the adjunctive-conjunction suffix *-te* 'and', which together surface as *=de* and conjunctively specify the conceptual domain in which the existence of the theme is located. The more colloquial form *=da* is a phonologically contracted — though pragmatically and informationally motivated — form corresponding to the more formal and literal *=de aru*.

Thus, through the above morphosyntactic and semantic analysis, it has been possible to account, by means of a single, consistent principle — namely, the **introduction of existence by [NP$_1$**

***ar-*]** and the **conjunctive specification of the domain of existence by [NP_2=*ni-te*], surfacing as [NP_2=*de*]**, schematically representable as **$E(NP_1) \wedge IN(rE(NP_1), NP_2)$, not reducible to $BE(NP_1, NP_2)$** — for:

- Morphosyntactic separability of *de* and *aru* in *=de aru*
- Syntactic and prosodic constituency of *=de* and the NPs preceding it
- Contextually optional omission of *aru* in *=de aru*
- Prosodic-word boundary between *=de* and *aru*
- Evidence for analysing *=de* as an adjunct-marking postposition–conjunction complex that functions as an eventuality-domain specifier $\wedge$ (r( ), NP)
- Foregrounding of the lexical meaning 'to exist' of the verb *aru*
- Grounds on which the Japanese desiderative *=de ari-tai* is not equivalent to the English *to want to be*
- Diachronic split from Old to Modern Japanese between *A B=ni ari → A B=nite ari → A=wa B=de aru* and *A B=ni ari → A=wa B=ni aru/iru*
- Evidence that the default negation of *A=wa B=de aru* is not the total *A=wa B=de nai* but the partial *A=wa B=de=wa nai*
- Syntactic and information-structural differences between *A=wa B=de aru* and *A=ga B=de aru*
- Underlying structure *=ni-te ar-0* surfacing as *=da*
- Pragmatic and informational motivation for the phonological reduction *aru → a* in *=da*
- Commonalities and differences between the so-called copular, nominal-adjectival and verbal-adjectival constructions

Put differently, to maintain the claim that *=da* is *be*, one would have to assume that the inherent meanings of *=ni*, *-te* and *ar-* — which together still autonomously underlie *=da* — have been eliminated or, alternatively, somehow changed into another meaning. One would then further need to explain when these meanings were eliminated or changed, why they were eliminated or changed, at what grammatical level they were eliminated or changed, and why meanings that are assumed to have been eliminated or changed nevertheless become foregrounded in desiderative constructions, exert structural effects under negation, create prosodic boundaries and maintain diachronic continuity.

Finally, while the present study reinterprets *=da* and *=de aru* as a postposition–conjunction–verb complex expressing E(A) ∧ IN(rE(A), B) rather than a copular predicate denoting BE(A, B), this proposal is not intended to invalidate previous research conducted within the copular framework. Much of the empirical and descriptive work on so-called copular sentences remains valuable and can be accommodated within the present account as analyses of the IN(rE(A), B) component, that is, as investigations into how the domain of existence is structured. In this respect, the present analysis may be regarded not as a rejection of the copular tradition but as an attempt to extend it by situating its insights within a broader model of Japanese predicational structure. Furthermore, the interaction between the existence-introducing component E(A) and the domain-specifying component IN(rE(A), B), which has received relatively little explicit attention in previous research, may provide a useful perspective for future studies on how these two semantic components relate to one another in information structure, negation and discourse.

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